EX NO: 28

SIMULATION OF DSSS COMMUNICATION SYSTEM WITH PN CODE GENERATION AND SIGNAL RECOVERY USING MATLAB

Objectives

i) To simulate a Direct Sequence Spread Spectrum (DSSS) communication system by generating a Pseudo Noise (PN) sequence and spreading the input data using MATLAB.

ii) To recover the transmitted data using the same PN sequence and analyze the effectiveness of the DSSS communication system.

CODE

```
clc;
clear;
close all;

% Data Bits
data = [1 0 1 1 0];

% Convert binary data to +1 and -1
data_bipolar = 2*data - 1;

% PN Sequence (Length = 5)
pn = [1 -1 1 -1 1];

% DSSS Spreading
spread = kron(data_bipolar, pn);

% Despreading
rec = reshape(spread, length(pn), []);
rec_data_bipolar = sign(sum(rec, 1));

% Convert back to binary
rec_data = (rec_data_bipolar + 1)/2;

% Display Results
disp('Original Data:');
disp(data);

disp('PN Sequence:');
disp(pn);

disp('DSSS Spread Signal:');
```

```
disp(spread);

disp('Recovered Data:');
disp(rec_data);

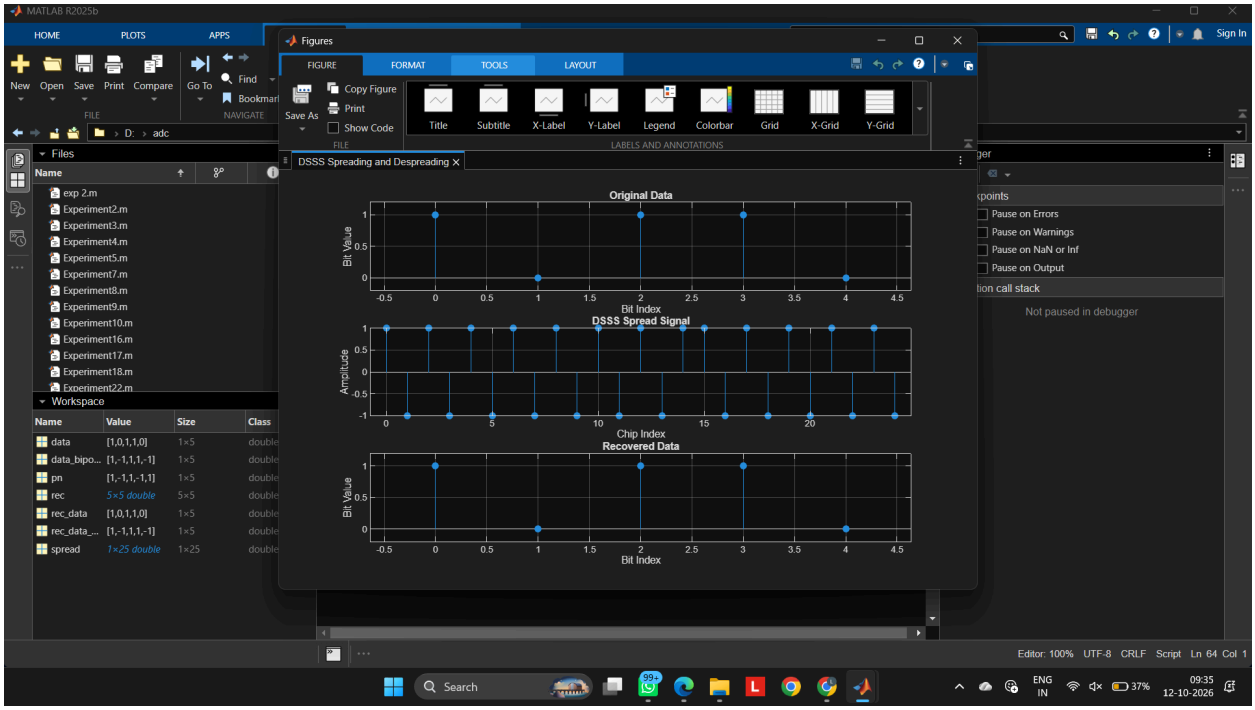
% Plots
figure('Name','DSSS Spreading and Despreading', ...
      'NumberTitle','off');

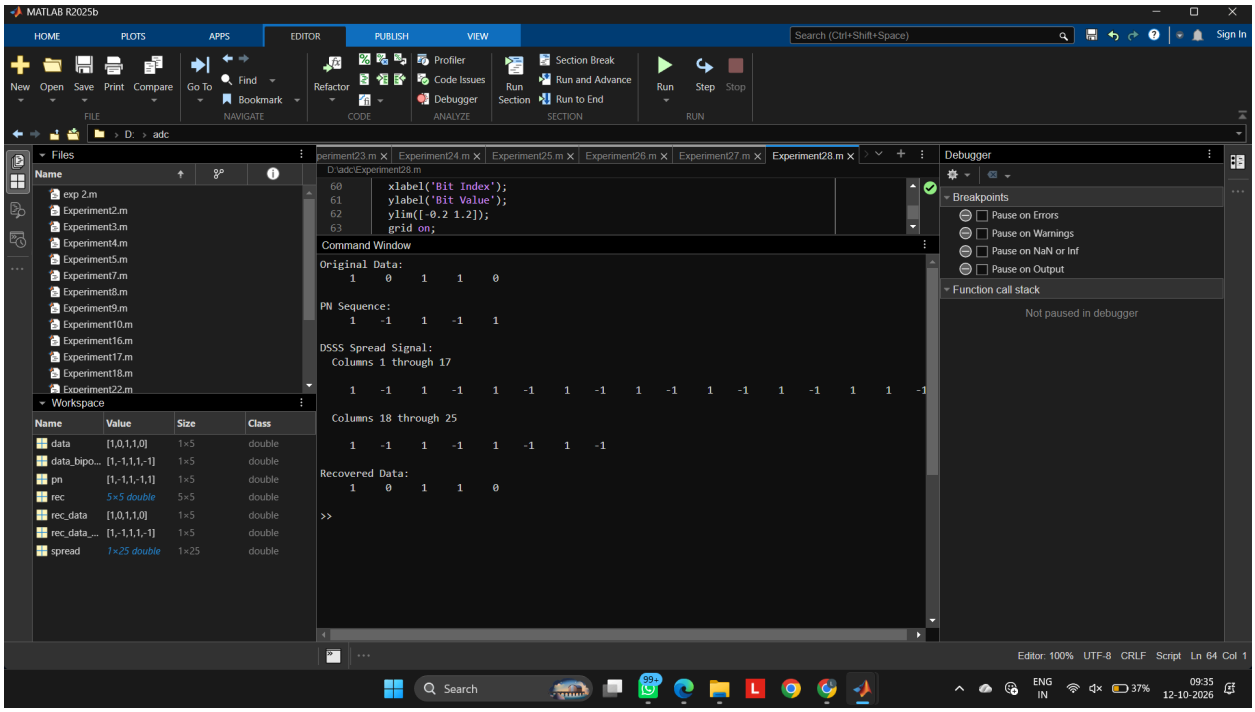
subplot(3,1,1);
stem(0:length(data)-1, data, 'filled');
title('Original Data');
xlabel('Bit Index');
ylabel('Bit Value');
ylim([-0.2 1.2]);
grid on;

subplot(3,1,2);
stem(0:length(spread)-1, spread, 'filled');
title('DSSS Spread Signal');
xlabel('Chip Index');
ylabel('Amplitude');
grid on;

subplot(3,1,3);
stem(0:length(rec_data)-1, rec_data, 'filled');
title('Recovered Data');
xlabel('Bit Index');
ylabel('Bit Value');
ylim([-0.2 1.2]);
grid on;
```

OUTPUT


Figures
DSSS Spreading and Despreading
Original Data
Bit Value
Bit Index
DSSS Spread Signal
Amplitude
Chip Index
Recovered Data
Bit Value
Bit Index


xlabel('Bit Index');
ylabel('Bit Value');
ylim([-0.2 1.2]);
grid on;
Command Window
Original Data:
1 0 1 1 0
PN Sequence:
1 -1 1 -1 1
DSSS Spread Signal:
Columns 1 through 17
1 -1 1 -1 1 -1 1 -1 1 -1 1 -1 1 -1 1 1 -1
Columns 18 through 25
1 -1 1 -1 1 -1 1 -1
Recovered Data:
1 0 1 1 0
>>